Review of Media Representation of Sport Concussion and Implications for Youth Sports

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KEYWORDS

- Sport-related concussion • Media • Removal from sport
- Chronic traumatic encephalopathy • Youth sport

KEY POINTS

- Many individuals learn about sport-related concussion (SRC) through various forms of media (newspaper, online news, social media, film, and so forth)
- Popular media does not always portray SRC accurately and there are many examples of misinformation in reporting.
- Knowledge about SRC varies considerably based on several factors, but accurate understanding of the injury does not always correlate to appropriate action.
- Given the high utility of various forms of media usage, this presents an opportunity to convey accurate SRC information.

INTRODUCTION

Participation in youth sport confers numerous mental health and physical benefits.[1] Nonetheless, participating in physical activity comes with inherent risks. Sport-related concussion (SRC), a type of mild traumatic brain injury (mTBI), is a risk associated with any physical activity. The impact of SRC and the potential long-term neurocognitive effects have been a mainstay in media over the past 2 decades. This has been driven by cases in which the postretirement declines of high-profile athletes have been attributed to SRC and repetitive brain impacts sustained during playing

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careers. Press coverage of these events has likely had a significant impact on public opinion regarding SRC and participation in youth sport. Rates of participation in some high school (HS) sports such as football and basketball have decreased in recent years for the first time in decades, notwithstanding the COVID-19 pandemic. Athlete and parent concern about SRC is frequently cited as a driving factor, and for those who do remain engaged in sport, fear of SRC remains high. Other beliefs about the short- and long-term impact of SRC are not always consistent with up-to-date science.[2] As varying forms of media remain major sources of SRC-related information, an exploration of its impact is worthwhile. Here, the authors examine some of the ways that media has represented SRC and how this representation may continue to influence youth sport participation today.

HISTORICAL PERSPECTIVE OF MEDIA AND SPORT-RELATED CONCUSSION

The first "Concussion Crisis" took place in the late nineteenth and early twentieth centuries. American Football was popular at that time, especially in the Ivy League colleges. The sport had few rules or regulations, and participants wore little protective equipment. Injuries abounded and team physicians were becoming increasingly aware of the hazards associated with play. The injury that we know today as concussion gained special notice. Popular media did not shy away from using colorful language to convey the risks of injury to their readers. In 1893, The New York Times noted that the sport incurred, "nearly the same risk that a soldier [assumes] on the battle field,"[3] and that, "an ordinary rebellion in the South American or Central American states is as child's play compared with the destructiveness of a day's game."[4]

Public alarm grew along with the understanding that SRC was a more nuanced and dangerous injury than previously thought. Physicians were beginning to appreciate that SRC might actually be difficult to detect and was occurring at a higher rate than previously believed.[5] Nichols and Smith noted that "A player might automatically run through a considerable series of plays before his fellows noticed that he was mentally irresponsible."[6]

Amid this fear and uncertainty, popular opinion called for change. In the 1900s, new rules made the game safer and it became more common for athletes to wear padding or other equipment.[7] Concerns persisted though, in particular regarding the long-term consequences of head injury. Harrison Martland published "Punch Drunk" in 1928 examining the long-term effect of head injury on boxers.[8] The public urgency to address the issue waned over the years, with much of the focus shifting to head injuries that were affecting the soldiers of World War I and II. It was not until the early twenty-first century that SRC was once again thrust into the spotlight.

Dr Bennett Omalu and colleagues reignited concern about SRC with his case reports on former National Football League (NFL) players with chronic traumatic encephalopathy (CTE).[9] In 2007, Boston University partnered with the Sports Legacy Institute to create the Boston University Center for the Study of Traumatic Encephalopathy, a group that has been on the frontline of concussion research since then. With this new wave of research, the popular media once again seized on SRC. The film *Concussion* was released in 2015, a semi-fictionalized take on Dr Omalu's research and the controversy surrounding the NFLs response. The film was criticized by much of the scientific community for oversimplifying the science and suggesting that football alone was the cause of death for athletes found to have CTE. Other contemporary headlines declared, "111 N F L. Brains, All But One Had CTE."[10,11] Recent publications have created an impression that nearly all professional football players will experience CTE, regardless of experiencing an identified concussion. Despite the tremendous

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amount of research that has taken place in recent years, many knowledge gaps persist, and media portrayal of scientific understanding of SRC is not always accurate.

CURRENT UNDERSTANDING OF SPORT-RELATED CONCUSSION: A BRIEF SUMMARY

An SRC is a type of mTBI induced by a biomechanical force that is transmitted to the brain. This impact leads to a constellation of neuropsychiatric symptoms (physical signs, behavioral, cognitive, or somatic changes) that typically resolve over the course of days or weeks. Recovery is characterized by an initial period of rest followed by a graded series of physical and cognitive steps that increase in difficulty and culminate in return to physical and academic activity.[12]

The impact that causes SRC leads to a cascade of events resulting in disruption of neurochemical homeostasis as well as increases in free radicals and other regulators of neuroinflammation. The energy required to reestablish homeostasis is believed to cause the hallmark symptoms seen during SRC recovery.[13] These changes take place on the microscopic level, and macroscopic changes (hemorrhage, fracture, gross lesions) are not consistent with SRC. Readily available neuroimaging (commuted tomography, MRI, x-ray) is unable to detect concussion and is used to rule out more severe injury.[14]

Concussion has historically accounted for 3% to 8% of all sport-related injuries that present to the emergency department.[15] Sports that involve athletes moving at a high velocity, interaction with an object (ball, puck, bat, and so forth), and/or intentional contact/tackling put participants at higher risk for injury.[16] Younger individuals may be more susceptible to SRC due to factors including mechanical features and being in a period of ongoing and rapid neurologic, cognitive, and emotional development.[17] However, Pierpoint and colleagues found that collegiate athletes had higher rates of SRC than HS athletes.[16] Overall, rates of concussion seem to be on the rise. This may be due to improved recognition and detection of SRC, improved awareness and appreciation of the importance of proper treatment and management, and changes in sport rules, equipment, and playing styles.[18]

The disease known as CTE is considered to be a result of repetitive concussions or sub-concussive impacts, the latter being a term with no consensus definition. It is diagnosed by pathologic findings of hyperphosphorylated tau proteins in perivascular locations at the depths of the cortical sulci.[19] These findings, only diagnosable on postmortem histopathological examination, are felt to be distinct from other neurodegenerative diseases. The number and severity of head impacts necessary to impart CTE has not been established. The degree to which CTE may cause neuropsychiatric symptoms, which are not required for diagnosis, is unknown. Although CTE has largely been studied in professional contact sport athletes and military personnel, it is likely present in many other individuals and the links to other factors such as age, medical issues, substance exposure, and genetics are unknown.[20]

Representation of Sport-Related Concussion in Popular Media

We sought to recreate the experience of a lay person looking to learn more about SRC. We used Google to search for "concussion in sport" and limited our results to material produced between January 2012 and December 2022 (**Fig. 1**). Browsing history was cleared to avoid any potential bias from past searches and we only used the News section of Google. The search resulted in approximately 116,000 articles. A random sampling from the first 10 pages (first 100 articles) resulted in 25 articles from reputable media outlets being selected. Five were excluded due to the requirement of a subscription or membership. Twenty articles were used for this examination (**Table 1**).

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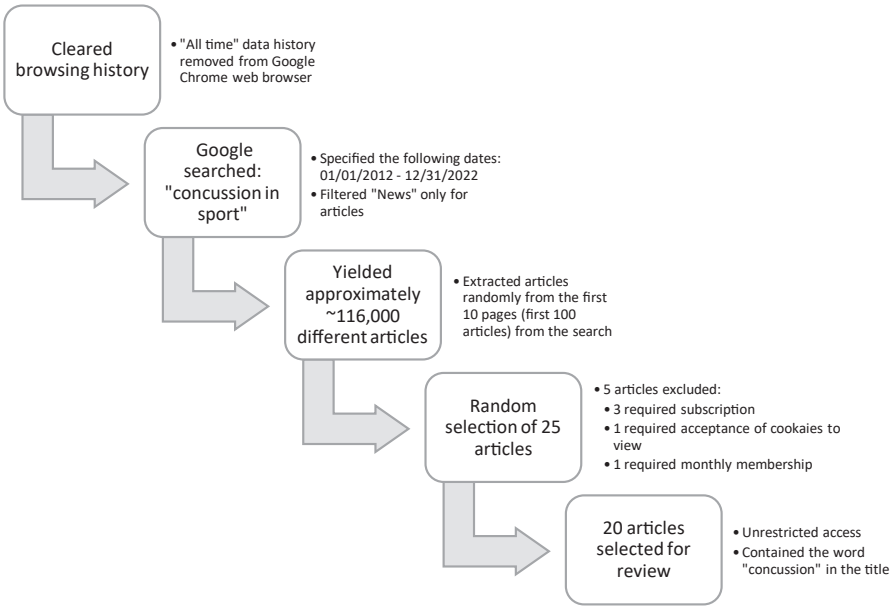

Fig. 1. Methodology for article analysis.

The term "concussion" was mentioned in all 20 articles. The terms "brain injury" appeared in 13/20 articles and "head injury" and/or "head impact" appeared in 13/20 articles. CTE appeared in 13/20 articles, and dementia was represented in 10/20 articles. CTE and dementia were typically mentioned as potential long-term risks associated with sustaining concussions or "repeated head injuries." The words "retirement," "retired," or "retire" were found in 9/20 articles. Both retirement and CTE/dementia reports were mostly seen after 2016 which correlates with several events covered in the media (NFL lawsuit and the movie *Concussion* in 2015). "Minor" or "mild" was used in 7/20 articles to discuss the severe brain injury. Other terms that were noted to recur: "subconcussive," "repetitive hits to the head," "repeat head trauma," and "cognitive decline." No idioms for SRC, such as "bell rung," were found in this sample, but "head clash" was used once in the sampled articles.

Our findings corroborate data found in several recent studies looking at the impact of social media has on SRC representation. The investigators from an analysis looking at mTBI-related press releases between January 2012 and December 2019 found a considerable amount of misrepresentation; 83% of 66 press releases reviewed exhibited at least one manifestation of "spin," of misleading titles (58%), reporting (74%), claims (67%), and inappropriate extrapolation (38%).[21] Another study revealed the frequent use of misleading terms reporting on SRC such as "mild concussion," "minor concussion," "slight concussion," and terms such as "concussion," "head injury," "brain injury," "head clash," and "brain trauma" being used interchangeably as descriptors.[22]

Eighty three percent of sports fans check a social media platform while watching an event on television.[23] A 2011 study (data collected for 1 week in 2010) revealed that Twitter is an active source for concussion-related information ranging from "news-related tweets" to "sharing personal information/situation" and "inferred management" of concussions.[24] Similarly, TikTok has gained significant traction since its inception in 2005. Carter and colleagues identified 157 TikTok videos that met inclusion criteria for concussion-related content—only two of the videos included

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Table 1
Data collection looking at the frequency of the below buzzwords related to concussion within each article (n = 20)

Media Source	Year	Concussion: Total[a]	Concussion: Filtered[b]	Head Injury and/or Head Impact	Brain Injury	Mild/Minor[c]	Chronic Traumatic Encephalopathy/CTE[d]	Retire, Retired, or Retirement[e]	Dementia
The Guardian	2022	22	13	1	0	0	7	0	0
CNN	2022	18	16	4	2	0	6	0	1
Futurity	2021	16	9	0	0	0	0	0	0
RTE	2020	24	18	5	4	3	14	1	5
ABC News	2019	19	10[f]	0 (“head clash” = 1)	0	0	3	1	0
ABC News	2019	40	28	4	0	1	1	0	0
Los Angeles Times	2019	18	15[f]	1	1	1	0	1	1
CNN Health	2019	46	34	0	2	0	0	0	1
ESPN	2019	45	38	4	1	0	1	1	0
News Medical Life Sciences	2019	24	20	14	0	0	7	3	2
NPR	2018	28	20	6	2	1	16	0	1
Healthline	2018	12	11	2	1	0	1	2	1
The Sport Journal	2016	81	60	1	1	0	8	3	1

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ARTICLE IN PRESS

Table 1
(*continued*)

Media Source	Year	Concussion: Total[a]	Concussion: Filtered[b]	Head Injury and/or Head Impact	Brain Injury	Mild/Minor[c]	Chronic Traumatic Encephalopathy/CTE[d]	Retire, Retired, or Retirement[e]	Dementia
The Sport Journal	2016	106	73[f]	25	2	1	3	0	1
The Conversation	2016	17	12	0	3	0	7	1	7
NPR	2016	31	20	0	3	1	0	0	0
Healthline	2015	24	17	2[f]	0	0	0	0	0
Bleacher Report	2015	12	9	0	3	0	0	0	0
National Academies	2013	43	39[f]	2	2	2	1	2	0
CNN Health	2013	22	17	0	0	0	0	0	0

[a] The number of times concussion appears on the Web site; inclusive of advertisements, picture headings, and title of article/subheadings.
[b] The number of times concussion appears within the body of the text; excluded the above from (1) and concussion pertaining to the title of the movie *Concussion or* any concussion work group or foundation.
[c] "Mild" or "minor" used to describe the severity of concussion or traumatic brain injury.
[d] Did not duplicate the count when this term was followed by chronic traumatic encephalopathy (CTE) as a descriptor. For example, CTE counted as 1.
[e] Articles described retirement as an outcome or referenced retired players in the context of concussion.
[f] Sport(s)-related concussion used at least once in the article.

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"educational content" with one being viewed greater than 1 million times.[25] Videos that came from a verified account tended to be the most viewed, and the vast majority of the videos reviewed took on a "light-hearted approach" to the topic of concussion.[25] Social media and personalized search engines use algorithms to increase the likelihood that users will continue to engage with their platform. Developers advertise stories or posts based off the relevancy of previous interactions. This can amplify risk of confirmation bias, especially when the content is scientific in nature. Post after post from similar sources or authors can potentiate the spread of misinformation and essentially change the algorithm for the user to remain in this perpetual cycle with limited interaction from opposing views or newer publications that may debunk past findings.

Media and Removal from Sport

Chris Borland was 24 year old when he retired from the NFL, citing "fear of concussions" as the primary factor for his decision. He had sustained an SRC in training camp yet kept playing immediately after and throughout the rest of the season. News outlets and articles referred to Borland as having experienced a "ding," "minor concussion," or his "bell rung." It was emphasized that he was in his first season as a linebacker when he made this decision to retire. He reported reading extensively about concussions and talking to teammates and family members about these fears. Articles quote him saying that reactions from his teammates were "mixed," some of whom cited money as a reason not to retire.

Borland is one example out of numerous who have retired from the NFL and expressed concerns about SRC and the long-term impacts.[26] Social media has been an integral source in spreading this information. Online responses have been mixed with some fans supporting Borland's decision, whereas others have suggested that he would be easy to replace.

This concern has been mirrored in youth sport populations. Rates of participation in youth sport have varied over the past decades, particularly in the past several years due to the COVID-19 pandemic. In 2021, 78% of children ages 6 to 12 years participated in a team or individual sport. However, only 37% regularly participated in a team sport on a regular basis, a decrease from previous years (45% in 2008). In the 13 to 17-year-old demographic, 74% participated in a team or individual sport, though only 42% participated on a regular basis.[27] American Football, lacrosse, ice hockey, and wrestling have shown persistent declines relative to a decade ago.[28] Early specialization, lack of access, time/financial costs, diminishing interest among participants, and concern about injury are all factors contributing to these shifts. Of note, participation in tackle football for children aged 6 to 12 year old declined 29% from 2016 to 2021, whereas participation in flag football increased 15%.[27]

Retirement from sport after SRC is complicated, with no consensus as to when this should be considered.[29] This decision should be highly individualized and take into account various factors including age, level of play, medical and psychiatric history, and family history. The biological, psychological, and social factors that influence a professional athlete to retire are considerably different than those impacting a youth athlete. The degree to which media representation of professional athletes retiring from sport may influence youth athletes or their parents to do the same is not yet known.

COLLEGE ATHLETE OPINIONS OF SPORT-RELATED CONCUSSION

Collegiate athletes have been found to have a relatively high level of understanding regarding SRC.[30] Attitudes are described as positive with athletes indicating that

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notifying medical staff and reporting injuries is important. Although contact/combat-sport athletes are more likely to receive education regarding SRC, they are less likely to report symptoms per one study, and data suggest that knowledge does not necessarily translate to positive reporting behaviors.[31] Underreporting behaviors are a significant concern with 15% to 45% of collegiate athletes electing not to report suspected or hypothetical concussive symptoms.[32,33] Across multiple studies, collegiate athletes often cited concussion nondisclosure to fear of missed playing time, fear of missing a game, fear of letting down teammates, or at the time, did not believe they suffered a concussion.[30–34] Athletes have cited medical professionals (73.2%) and school-based professionals (68.4%) as sources of SRC education, and nearly one-third cited books, movies, social media, or sporting news outlets as an SRC resource.[32]

HIGH SCHOOL AND YOUTH ATHLETE OPINIONS OF SPORT-RELATED CONCUSSION

Although youth HS athletes may possess some accurate knowledge regarding SRC, there remain significant gaps in understanding and application of this knowledge. General SRC knowledge and attitudes are variable with SRC knowledge ranging widely (39%–80%).[35–40] One study found that 12% of youth English rugby players adhered to the recommended return to play (RTP) guidelines.[37] Miyashita and colleagues surveyed a variety of HS athletes and found that 95.6% agreed to the statement that "there are long term effects" from SRC, yet 50.9% also described believing that "the importance of the game should affect RTP decisions."[35] A study of New Zealand rugby players found considerable differences in SRC knowledge and attitudes with similar declines in reporting behaviors toward the end of season when competition needs were higher.[36]

Doucette and colleagues found that female athletes had higher SRC knowledge scores, and soccer players scored higher than collision athletes.[40] Athletes from in lower socioeconomic areas scored lower in knowledge and attitudes toward SRC.[36] Bloodgood and colleagues surveyed 252 HS athletes with most viewing SRC as a "critical issue." The younger athletes were more likely to believe concussions were a "critical issue" compared with the older athletes of this study (ages 16–18 years; 34%). More than half of the youth athletes disagreed with the statement "I am fearful that my circle of friends would think I was dumb for caring about concussion."[41] Approximately half of HS athletes would not report SRC symptoms.[40,42] A 2020 systematic review of SRC reporting behaviors found evidence for underreporting behaviors in all 26 studies.[43] Reasons for underreporting included: "not wanting to leave the game, not thinking their injury was serious enough, and wanting to avoid letting the team down." High levels of concussion knowledge (>80% correct on concussion fact knowledge) did not correlate with increased reporting of SRC, similar to that of collegiate athletes in other studies.[34,43]

Seventy one percent of New Zealand soccer players identified their coaches as a primary source on SRC information.[36] Youth athlete concussion knowledge was positively influenced by higher parental SRC knowledge, history of prior SRC, and older athlete age.[44] Most of the Canadian HS athletes identified their parents or guardians as a primary source of SRC information,[45] whereas 22% of South Korean athletes had heard of SRC from media.[46]

Parental Opinion of Sport-Related Concussion

Adults displayed high knowledge regarding concussions, often scoring ≥75% on SRC-related questionnaires.[38,47] Despite this, parents frequently answered incorrectly

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believing that neuroimaging studies are capable of detecting SRC.[38] Parents also tend to be concerned about permanent brain damage due to SRC.[47] Ninety-six percent of parents reported belief that there are long-term effects of SRC.[48] Parents of children who play contact sports (37%) are more likely to be concerned their child will develop CTE as compared with those who do not (21%).[49] In comparison surveys, parents were more likely to express concern for long-term consequences of SRC, whereas the youth athletes were more concerned about RTP.[50] Kroshus and colleagues found that more than half of parents were somewhat or very worried about concussion.[51] Nearly 87% reported worry about injury risk with concussion as the injury cited most according to a recent survey.[27] Parents acquire their information from a variety of sources, the primary of which are doctors (35%) and popular media (12.8%). Other studies have found that nearly half of adults obtain SRC information from news media, 36% from sports media, and 34% from physicians.[38]

Fear of SRC and potential long-term sequelae is a common driver for voluntary removal from sport. Hibshman and colleagues found that 49% of parents would prohibit their children from participation in a collision sport, primarily listing safety concerns.[52] Twenty-six percent of parents that previously allowed their child to play football indicated that they would no longer allow them to continue.[38] Nearly half of American parents reported that they would encourage their child to play a sport other than tackle football due to fear of concussion, up 8% from 4 years prior.[53] There are adult factors that support youth contact sports participation including being a female parent, older parent, having an older eldest child, or personal prior participation in contact sports.[39] Adults (with and without children) were less likely to allow daughters to participate in contact sports.[39]

DISCUSSION

Over the past several decades, both youth athletes and parents have become keenly aware of SRC and the potential impact the injury can have. Researchers continue to expand our understanding of SRC, exploring the factors that can predict injury, detection and diagnosis, recovery, treatment, long-term impact, and the links between head impacts and CTE. Despite ongoing questions in the medical community, many lay individuals are under the impression that science has established a consensus answer for these issues. A likely factor for this is how these issues are being reported. In addition to medical professionals, popular media, including entertainment and social media, is the most common source of information for many individuals, with certain demographic factors influencing this. Communities with less access to medical professionals may fill that void with media as their primary source for SRC guidance. This can be problematic, as popular media has been demonstrated to frequently misrepresent SRC data.

Misinformation, spin, omissions, or inaccurate conclusions are unfortunately common issues.[21] Measured and evidence-based reporting of credible research may run secondary to attention grabbing headlines. Our media search found that CTE was mentioned in 65% of articles reviewed and retirement/retired athlete in 50%. Only 25% did not contain a mention of CTE, dementia, or retirement, and those articles tended to be older. If a parent or athlete was to have a similar search experience, it is likely that the idea of CTE and retirement would be suggested to them. Many individuals do not have the scientific background to ascertain whether their risk for these issues is significant or indicates a change in their activity or sport of interest.

Other headlines might have a similar anxiety-inducing effect. The media portrayal of the work done by Mez and colleagues is a notable example.[11] In a paper supplied by a

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convenience sample of brains from former football players donated for CTE research, they found that a high percentage had the neuropathological findings of CTE. A New York Times article later reported that all but one of the brains from 111 former NFL players had CTE. A casual reader might easily interpret this to mean that participation in a high level of football is all but a guarantee to develop CTE, something that cannot be established based on the data presented in that study. The biological, psychological, and social factors that so often impact retired athletes are often minimized. Retired athletes can be subject to financial, personal, social, and injury/pain medication-related stressors, all of which can mimic the behavioral and cognitive features associated with CTE. Factors like those rarely impact amateur or youth athletes. Reporting, including all these issues, may shift parental concerns about participation in youth sport.

Journalists can also ensure the accurate use of concussion-related terminology. Descriptors such as mild, moderate, or severe are inaccurate, and colloquialisms such as a "ding" or having ones "bell rung" should be avoided.[22] When reporting on the personal experiences of concussed athletes, factors such as level of play, biological and medical attributes, and social circumstances can be mentioned to avoid extrapolating their decisions or the risks that they face onto other individuals.

Mass media and news communications play a pivotal role in the dissemination of public health information. Popular media is intentionally forming not only what is shared, but the context in which information is shared and distributed to consumers. Regardless of scientific accuracy, this material has the potential to reach millions of people and influence public perception of a topic almost instantaneously. There is a tremendous amount of potential that various social media platforms have when it comes to using entities (ie, National Collegiate Athletic Association [NCAA] and NFL) or "influencers" such as pro-athletes, to help promote concussion awareness and management and to stop the spread of misinformation. Media also has the power to educate a wide audience, which can positively influence concussion awareness and knowledge through factual reporting and advocacy.

Another concerning factor is that an accurate understanding of SRC does not always lead to appropriate response. Despite appropriate SRC knowledge, many youth athletes admit that they would not report concussion symptoms, or feel that preemptive return to activity would be appropriate given an important enough game. Youth athletes will often cite fear of missing playing time or letting down their coaches as major contributors as to why they would not report their injury. Another theory is that appropriate SRC education may in effect "coach" athletes as to how to hide symptoms or feign recovery, a response that could be disastrous.[54] Athletes do get a great deal of SRC information from news and social media, presenting an opportunity to shine a focus on the need for immediate and accurate reporting of symptoms after injury.

Parental anxiety regarding risk of SRC should be measured against the countless benefits of participation in sport. In particular, team sports are proved to increase confidence, decrease depression and anxiety, and enhance feelings of interconnectedness with peers, in addition to the physical benefits of activity.[1] Although no one should be encouraged to engage in any sport that they do not feel safe in, a consideration of the risks and benefits, as well as safe alternatives, should be given. Rarely would complete removal from all sport be necessary.

SUMMARY

The risk of SRC is real and should not be taken lightly. However, it should be emphasized that with appropriate treatment, this is an injury from which recovery is very

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likely. Further research regarding the long-term risks of participation in sport and repetitive head injury is needed. The benefits of sport participation should be measured against the risks, and every effort should be made to make sport accessible, equitable, and safe for all involved to maximize the psychosocial benefits of sport. To that end, media representation should emphasize an accurate representation of the current consensus of SRC. In doing so, we can hope to equip athletes, parents, and coaches with the appropriate information with which to make important decisions about participating in sport.

CLINICS CARE POINTS

- Clinicians should be aware of where their patients acquire sport-related concussion (SRC) information from, misunderstandings they may have, and provide appropriate education regarding this injury.
- Ensure that your patients and athletes have an accurate understanding of SRC and appropriate responsiveness to injury.
- Stress the importance of a full recovery and return to baseline before full return to sport.
- Participation in youth sport confers many benefits such as improved confidence and feelings of interconnectedness, and the inherent risks associated with play should be weighed against the benefits.

DISCLOSURE

The authors have nothing to disclose.

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